

Data-driven evaluation of existing bridges: Methods and applications at ZAG



TECHNICAL WORKSHOP



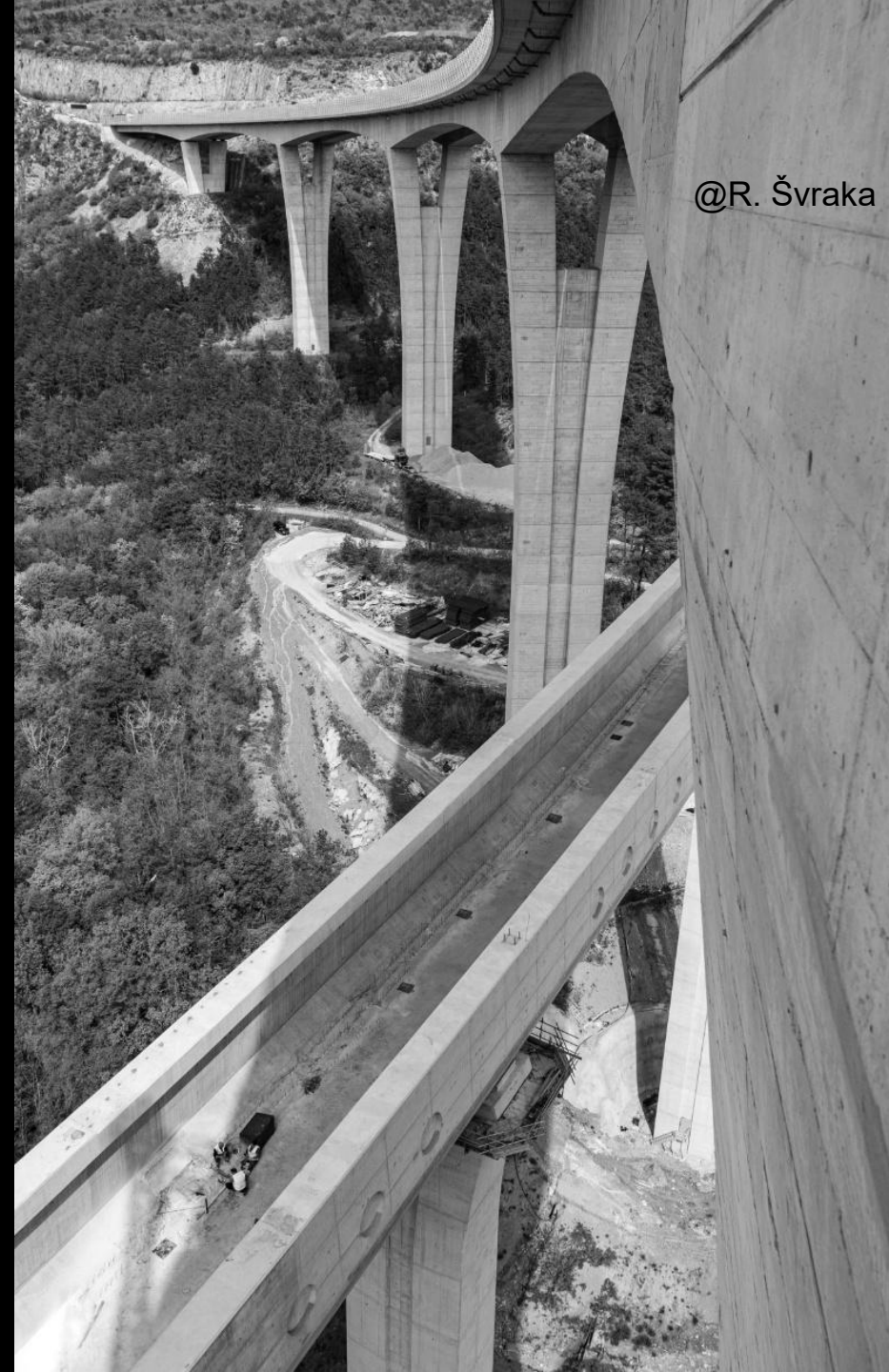
University of
Southampton

CIVIL, MARITIME AND ENVIRONMENTAL
ENGINEERING (CMEE)

ANDREJ ANŽLIN

Section for bridges and Other Engineering Structures
Department of Structures

23. 9. 2026



@R. Švraka

...bridges are stronger than we think...

Load effects are less than in the codes!

Despite being deteriorated, bridges are likely safe, but...

... how to prove their actual safety?



ZAG BRIDGE ASSESSMENT SCHOOL

- strain based, 30 yrs, USA, traffic load focused

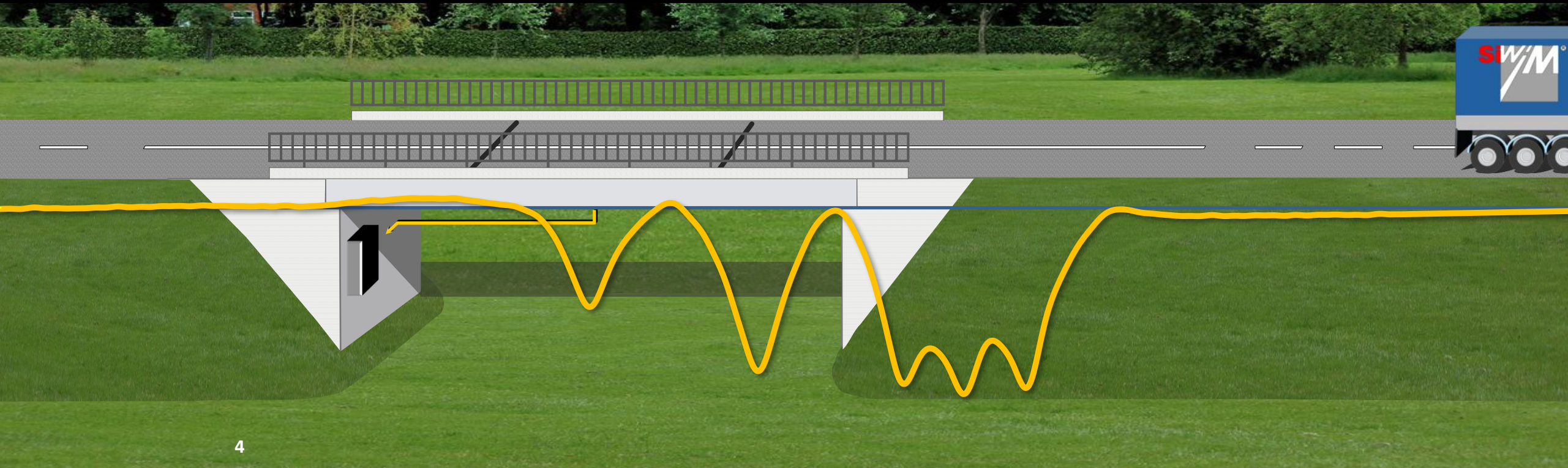
$$\textit{Rating Factor} = \frac{\overset{\text{capacity}}{\Phi \cdot R_d} - \overset{\text{dead load}}{\gamma_D \cdot G_D}}{\underset{\text{traffic load}}{\gamma_L \cdot G_L} \cdot DAF}$$

... highly based on Bridge Weigh-in-Motion system

Bridge Weigh-in-Motion system (B-WIM)

ZAG

Turning bridges into scales for moving traffic.

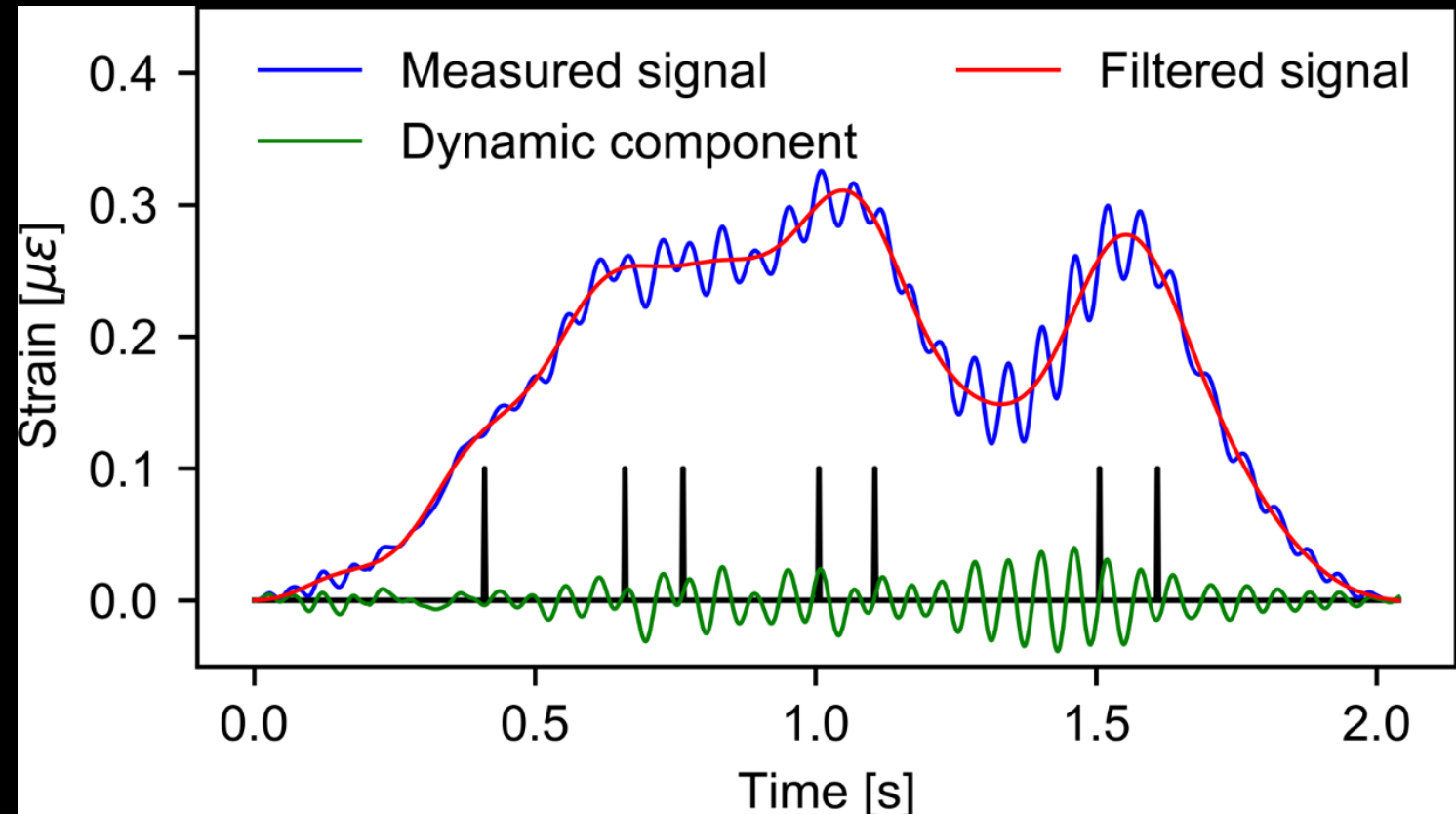


4

Axle load & spacing, Influence line, GDF, DAF, SHM extensions

Dynamic amplification factor (DAF)

$$DAF = \frac{S_{total}}{S_{static}}$$



Measurements of bridge dynamic amplification factor using bridge weigh-in-motion data

Jan Kalin , Aleš Žnidarič , Andrej Anžlin  and Maja Kreslin

Slovenian National Building and Civil Engineering Institute, Ljubljana, Slovenia

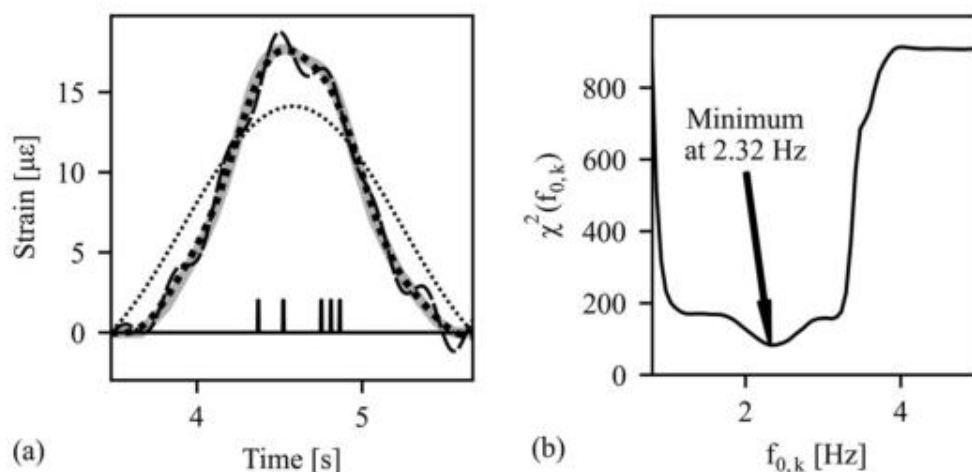


Figure 5. Example of filtering with different cut-off frequencies (a) and the minimised function (b).

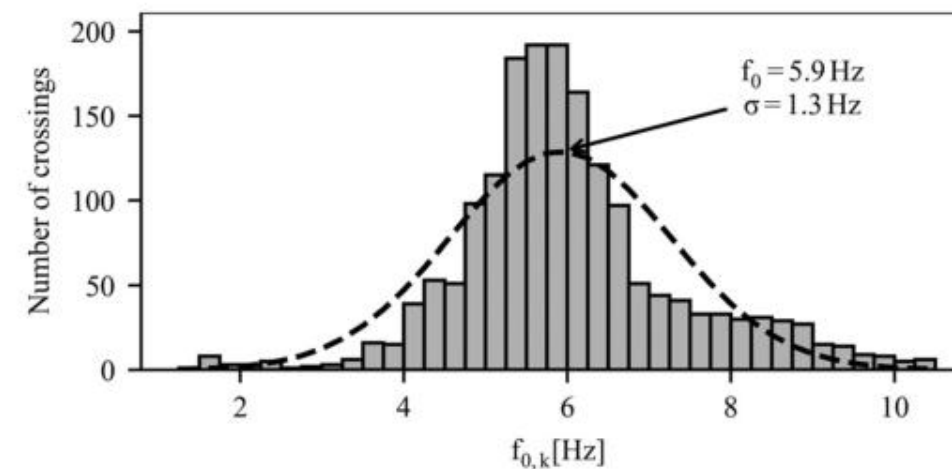


Figure 6. The histogram of cut-off frequencies and its associated Gaussian distribution for site SI05.

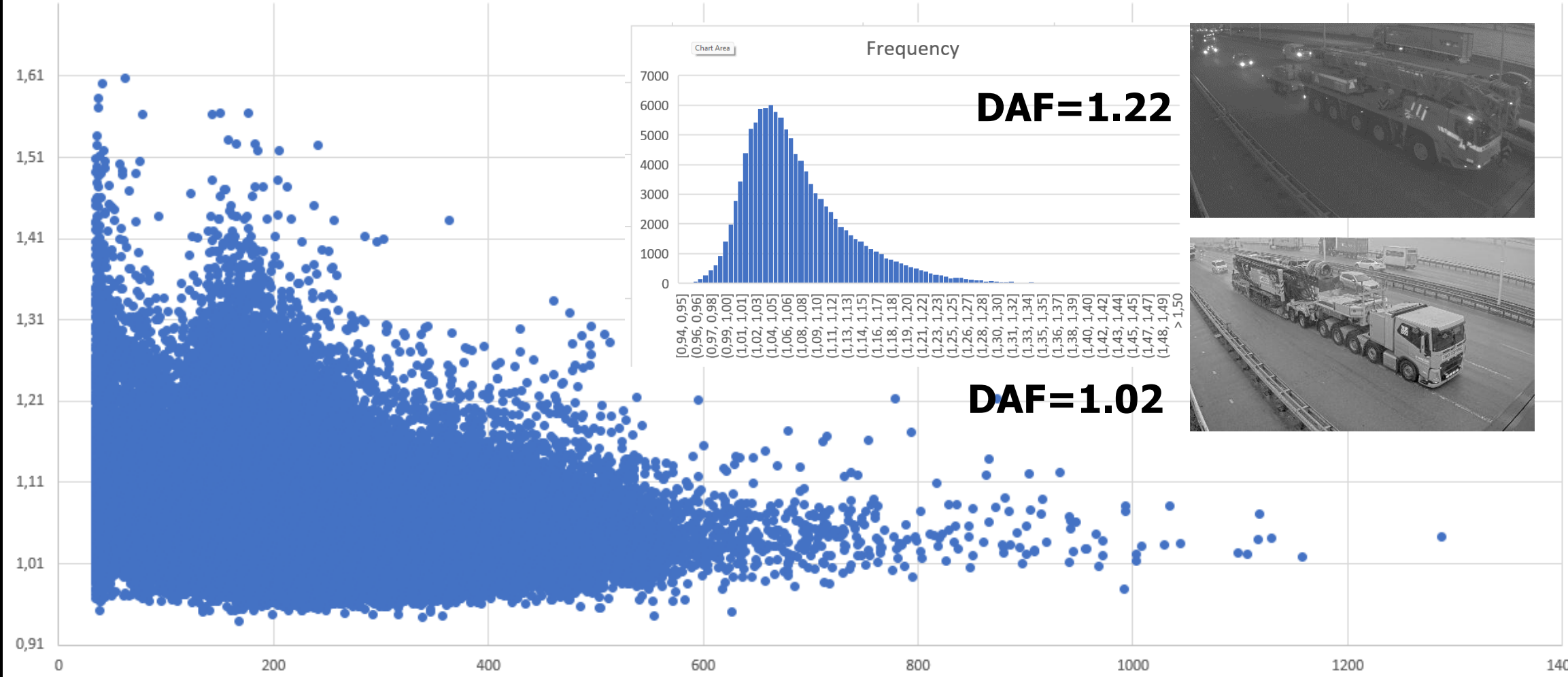
DAF | 2 weeks | >100.000

DAF=1.53



G

DAF



DAF=1.22



DAF=1.02



GVW (kN)

**DETAILS ARE
IMPORTANT**

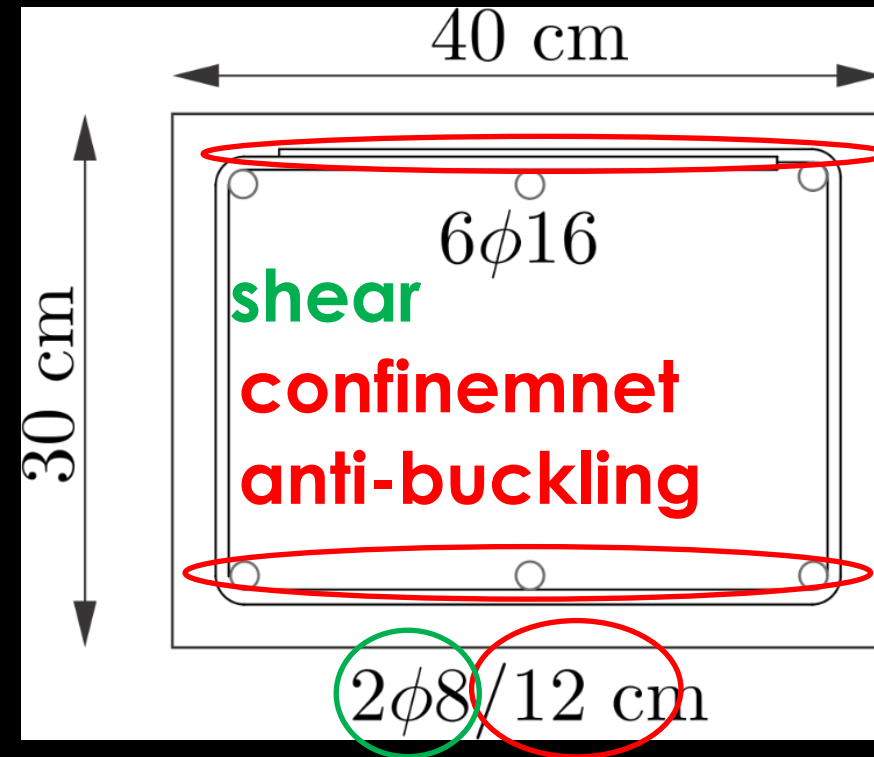


INADEQUATE DETAILS

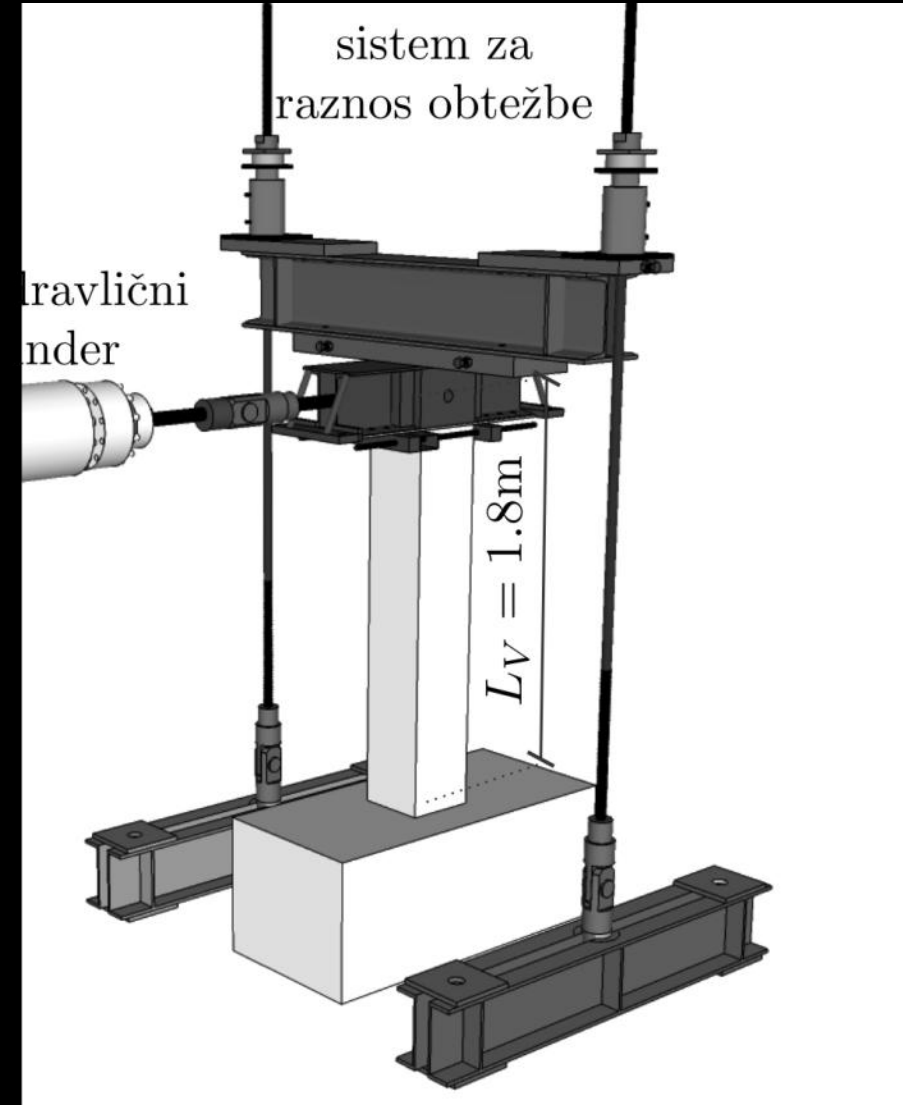
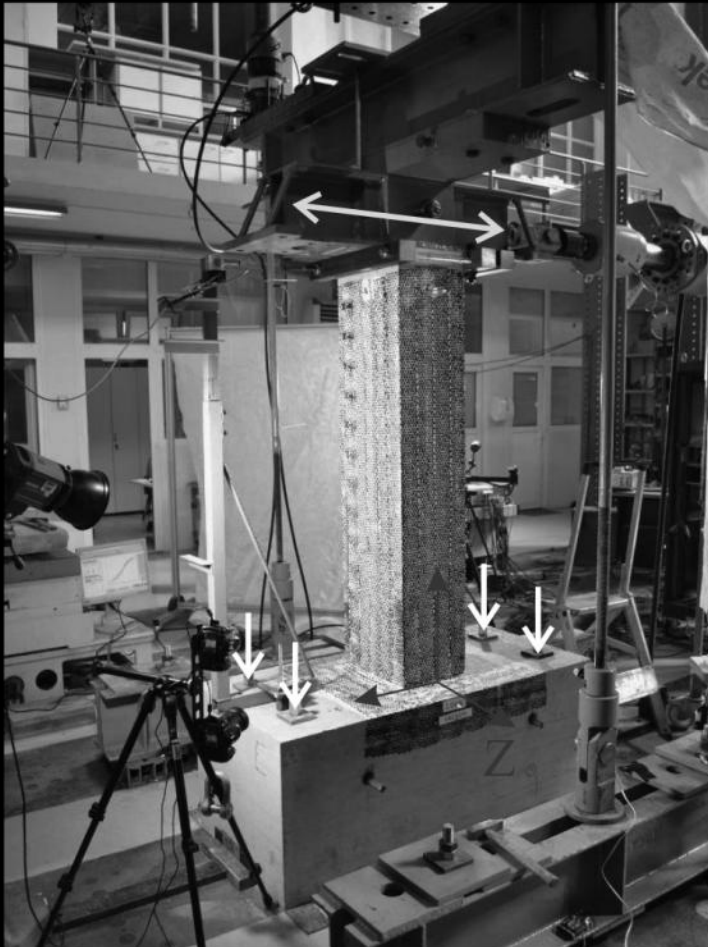


Local buckling

Global buckling



CYCLIC TEST OF RECTANGULAR RC COLUMNS



CYCLIC TEST OF RECTANGULAR COLUMNS

STD/135



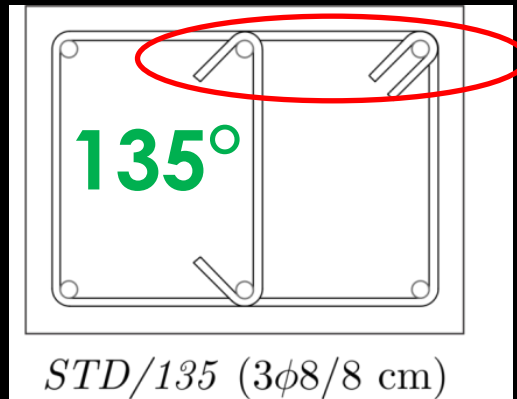
Phase #	Rotation Amplitude [%]
10	4.0
11	5.2
12	6.4
13	7.6
14	8.8

STD/90

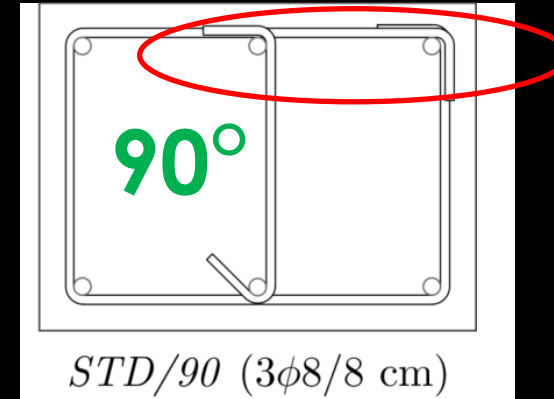


CYCLIC TEST OF RECTANGULAR COLUMNS

Different details

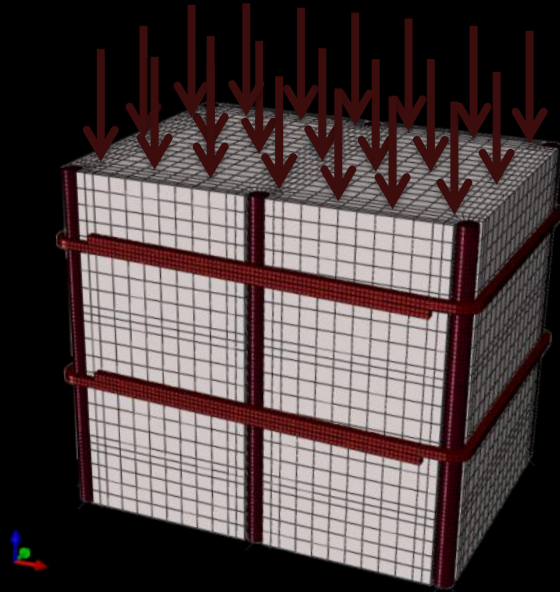


$$\theta_{NC} = 7,5\%$$

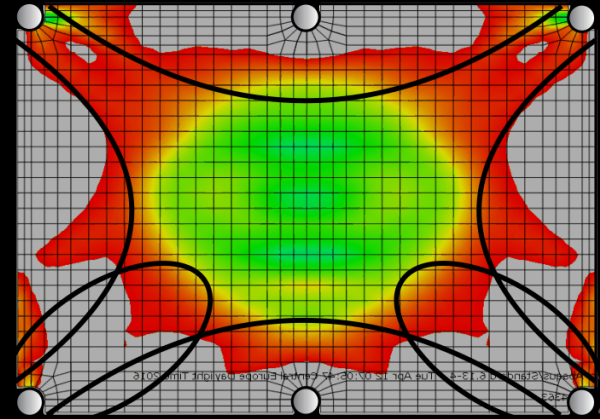


$$\theta_{NC} = 6,9\%$$

MICRO MODELLING OF CONCRETE CONFINEMENT



Arch effect
 $\alpha \neq 45^\circ$



Areas of poorly
supported bars

CASE STUDIES

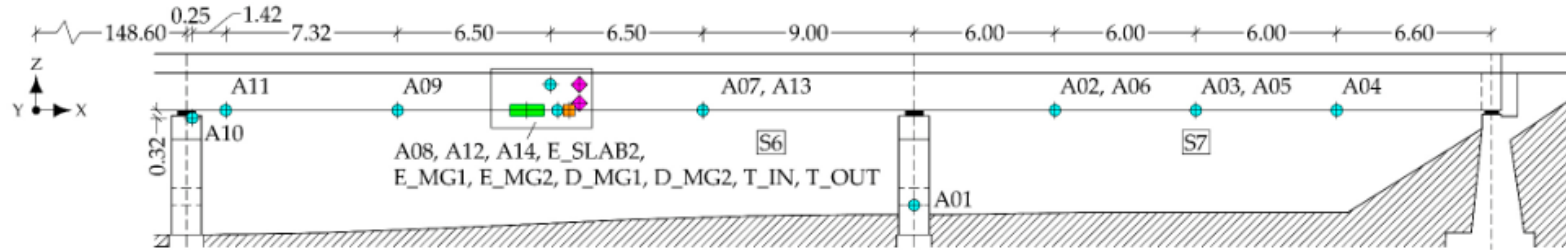
Tomačevo case study

- Built in 1982
- Seven-span continuous post-tensioned double-T section superstructure
- Two parallel structure
- Cast in place
- Minor rehabilitation in 1990 and 2018

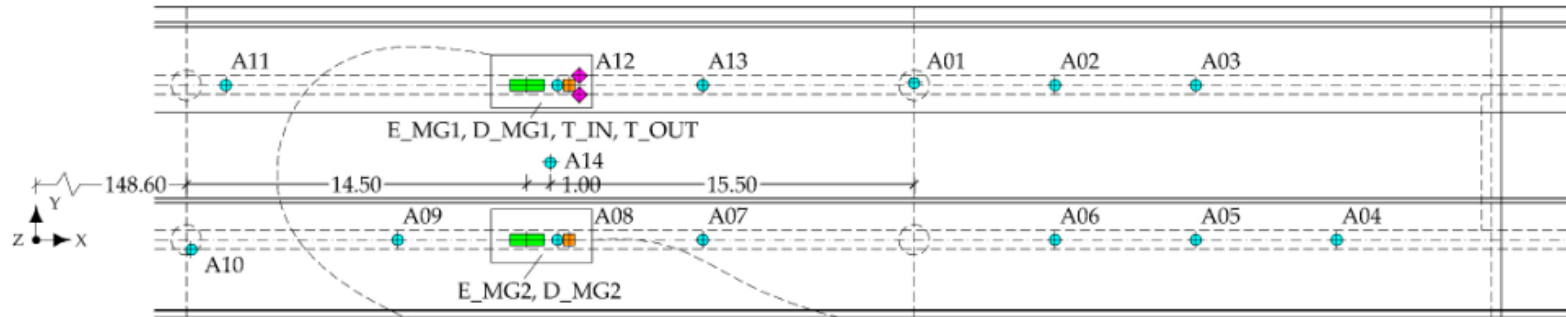


Sensor disposition (B-WIM + SHM)

SIDE VIEW ALONG UPSTREAM EDGE WITH SENSOR DISPOSITION: SPANS S6 & S7

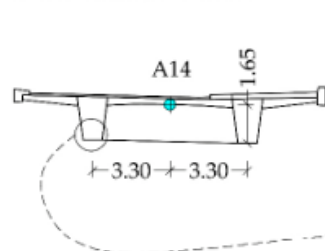


PLAN VIEW WITH SENSOR DISPOSITION: SPANS S6 & S7

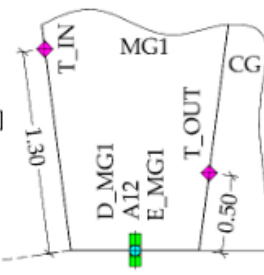


DETAILED VIEWS NEAR MID-SPAN

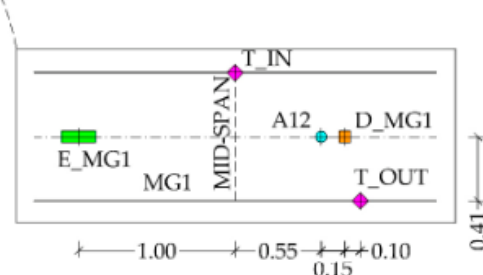
POSITION OF A14 IN CROSS-SECTION VIEW



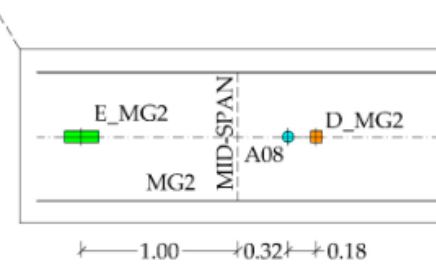
MG1: CROSS SECTION



MG1: PLAN VIEW



MG2: PLAN VIEW



$$\begin{aligned}\Sigma_{\text{ACC}} &= 14 \\ \Sigma_{\text{STRAIN}} &= 2 \\ \Sigma_{\text{DISPL}} &= 2 \\ \Sigma_{\text{TEMP}} &= 2\end{aligned}$$

+ B-WIM

Calibration vehicle passages



(a)



(b)

Figure 7. Calibration vehicles: V1 (a) and V2 (b).

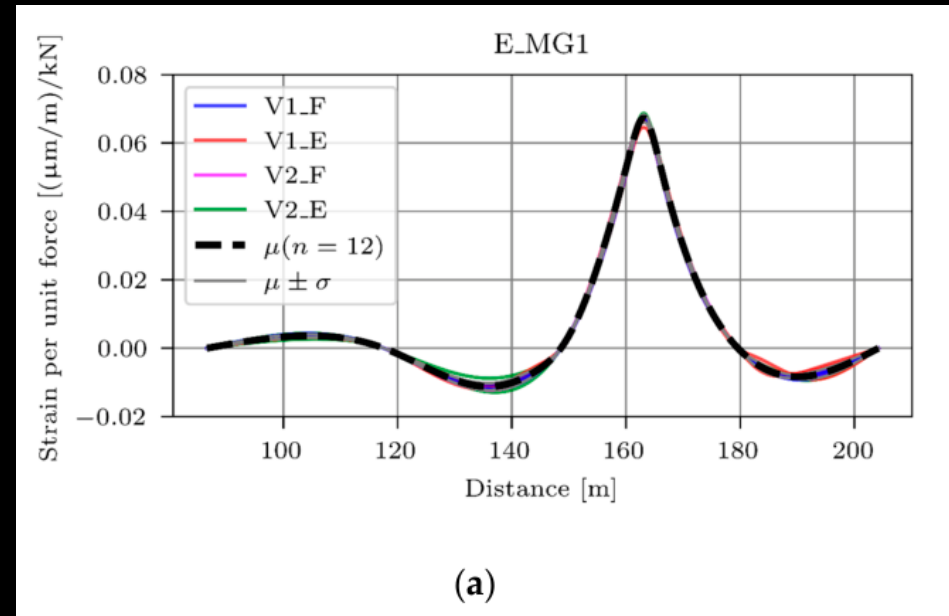
Table 1. Axle loads, axle spacing, and gross vehicle weights (GVW) of the calibration vehicles.

	1st Axle		2nd Axle		3rd Axle		4th Axle		5th Axle	
Vehicle ¹	Load (kN)	Spacing (m)	Load (kN)	Spacing (m)	Load (kN)	Spacing (m)	Load (kN)	Spacing (m)	Load (kN)	GVW (kN)
V1_F	68.67	3.42	73.58	1.37	71.61	/	/	/	/	213.86
V1_E	57.88	3.42	37.28	1.37	36.30	/	/	/	/	131.45
V2_F	69.65	3.60	90.25	2.82	64.75	1.31	61.80	1.31	59.84	346.29
V2_E	56.90	3.60	29.43	2.82	14.72	1.31	15.70	1.31	14.72	131.45

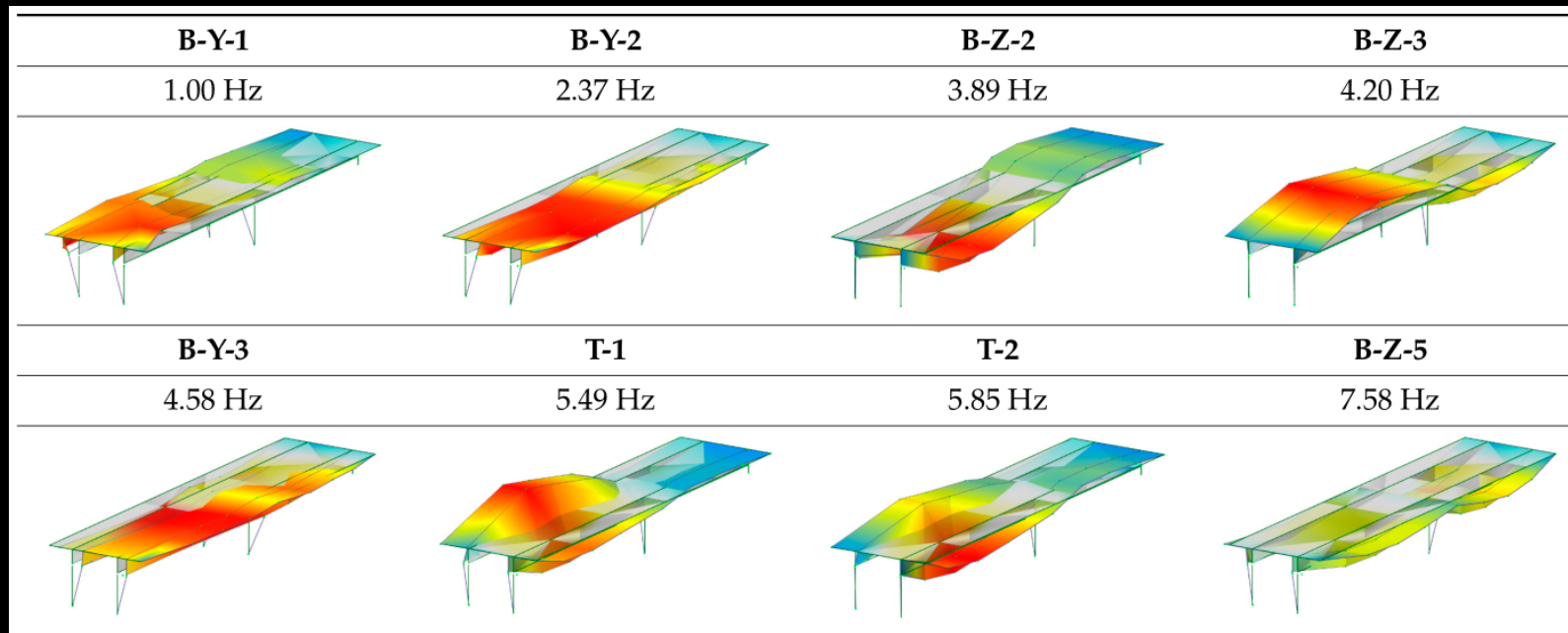
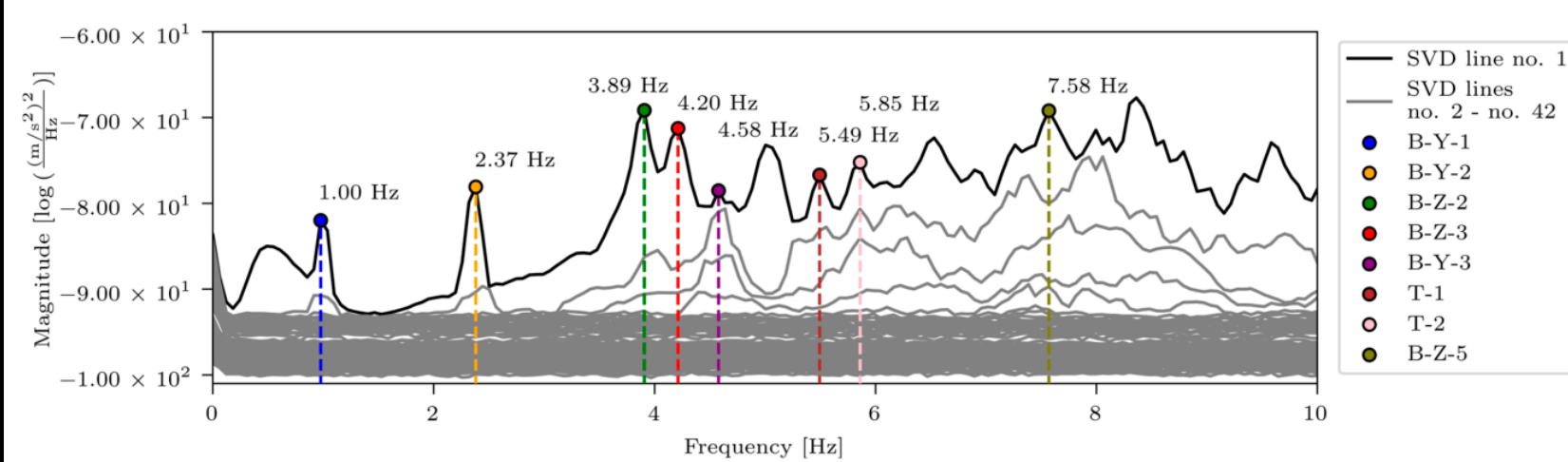
¹ F represents a fully loaded truck, and E an empty truck.

$\Sigma = 72$
(2 NIGHTS)

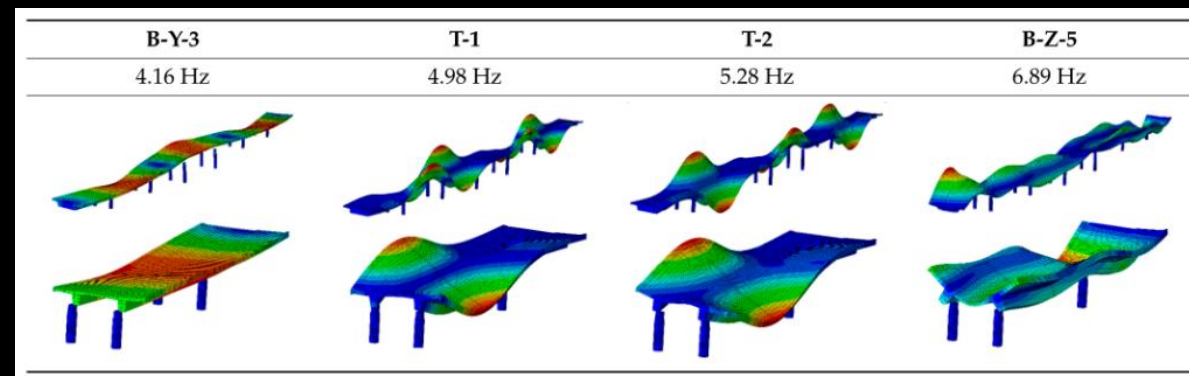
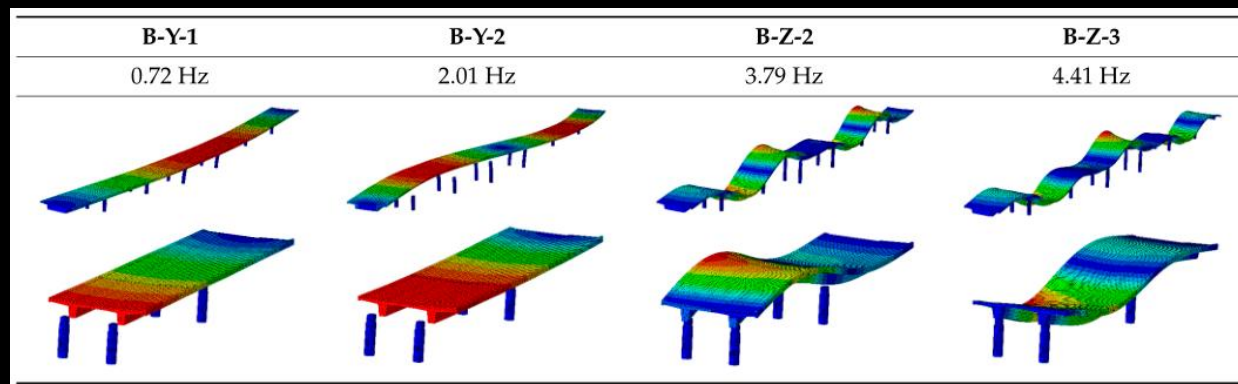
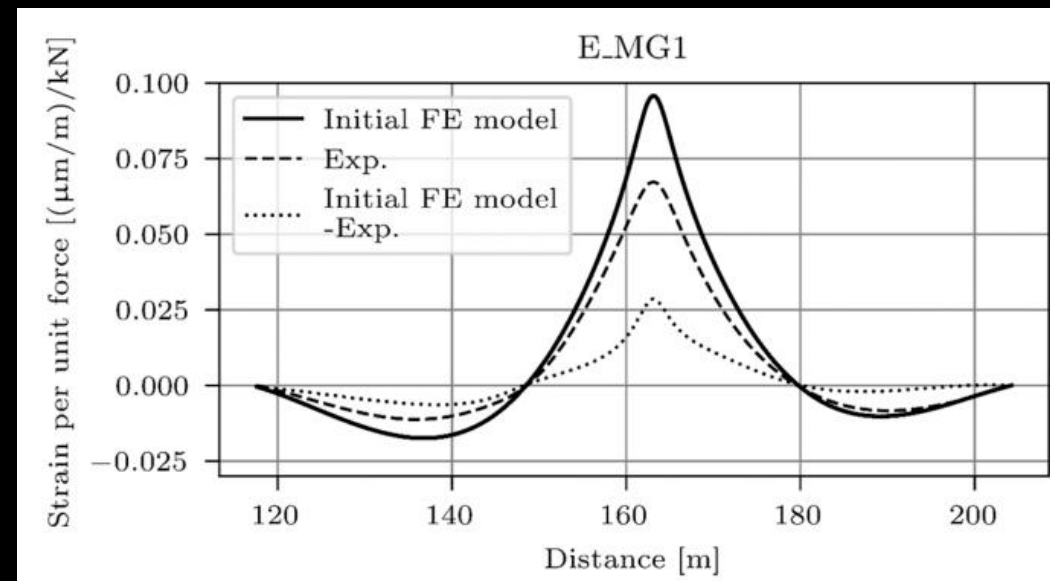
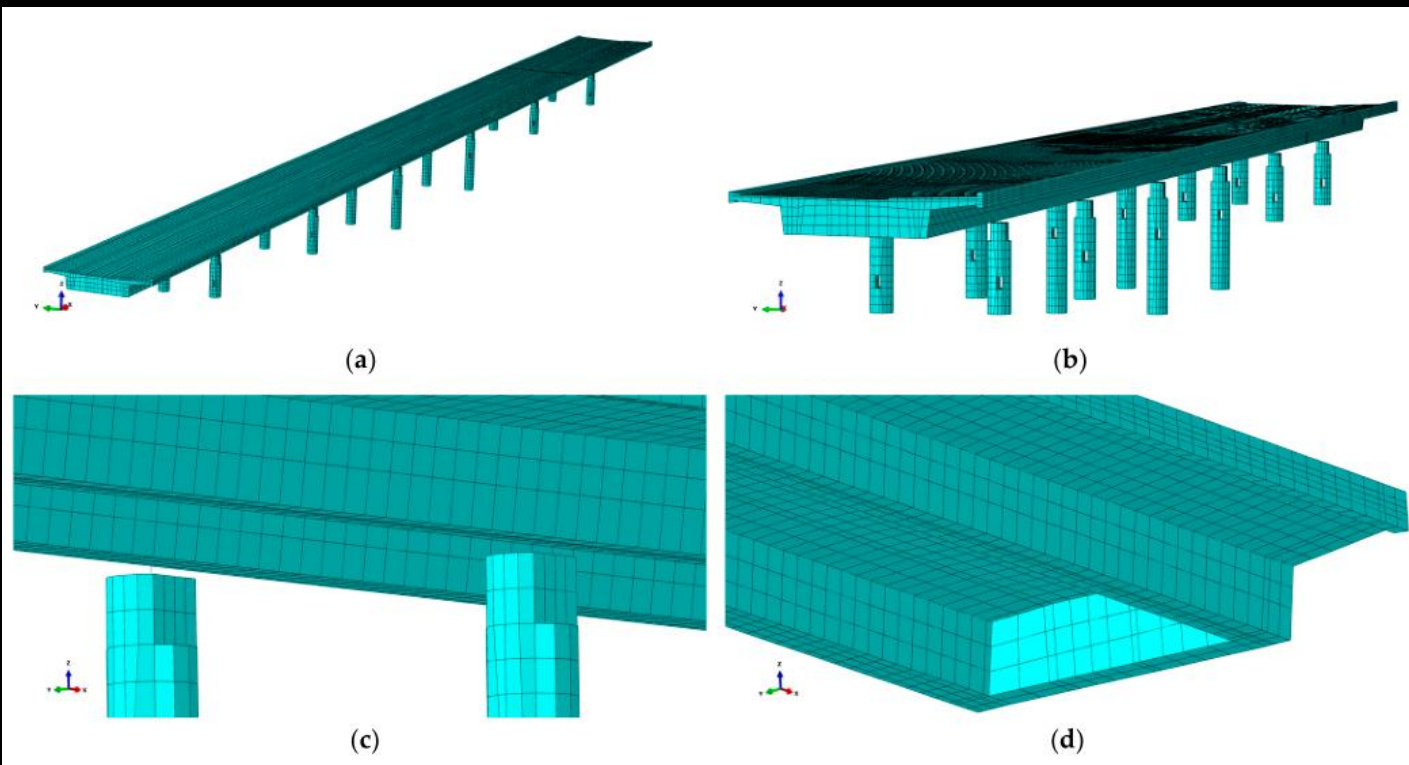
Extracted influence lines (from B-WIM)



Measured modal parameters

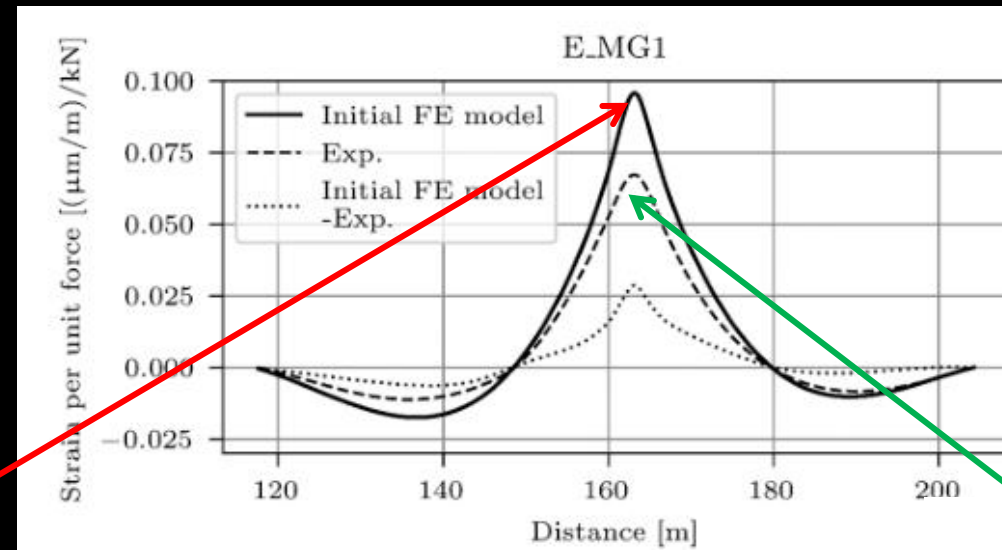


Finite element (FE) model



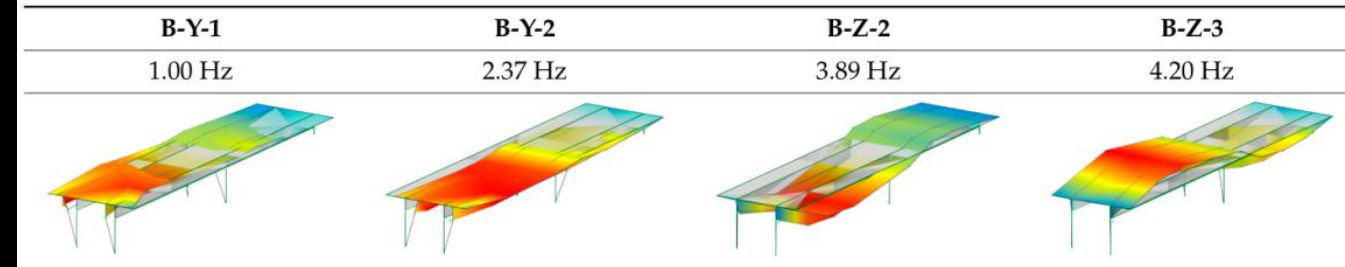
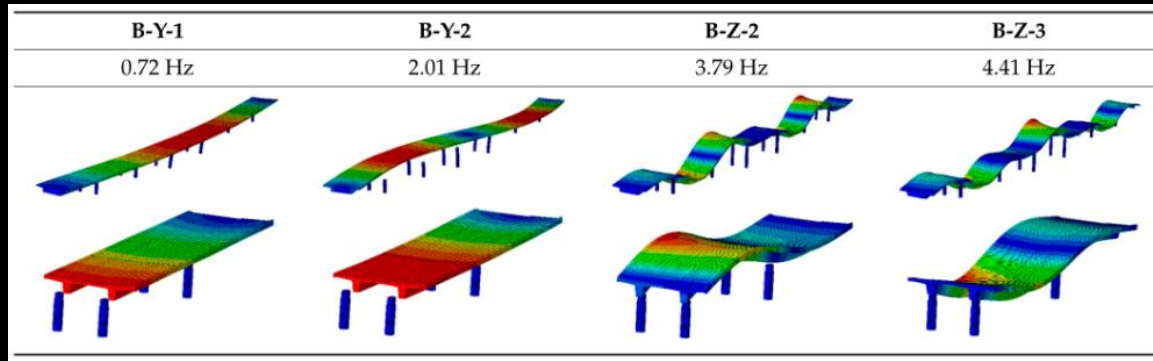
Finite element model updating (FEMU)

- Finite element (FE) models are updated to match the **analytical** and **measured** responses.
- Error-Domain Model Falsification



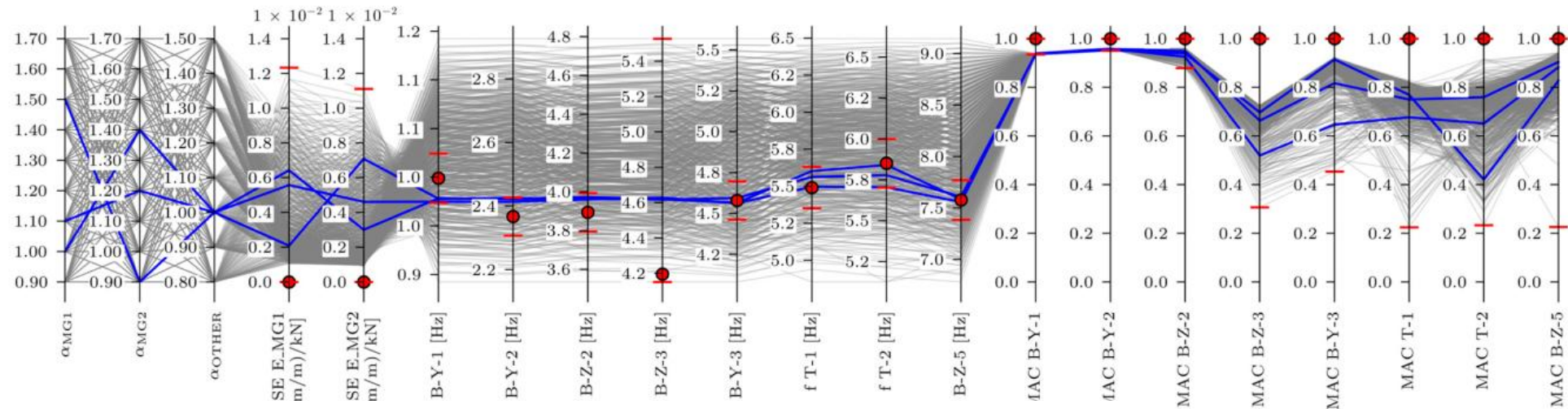
MODELLED

MEASURED

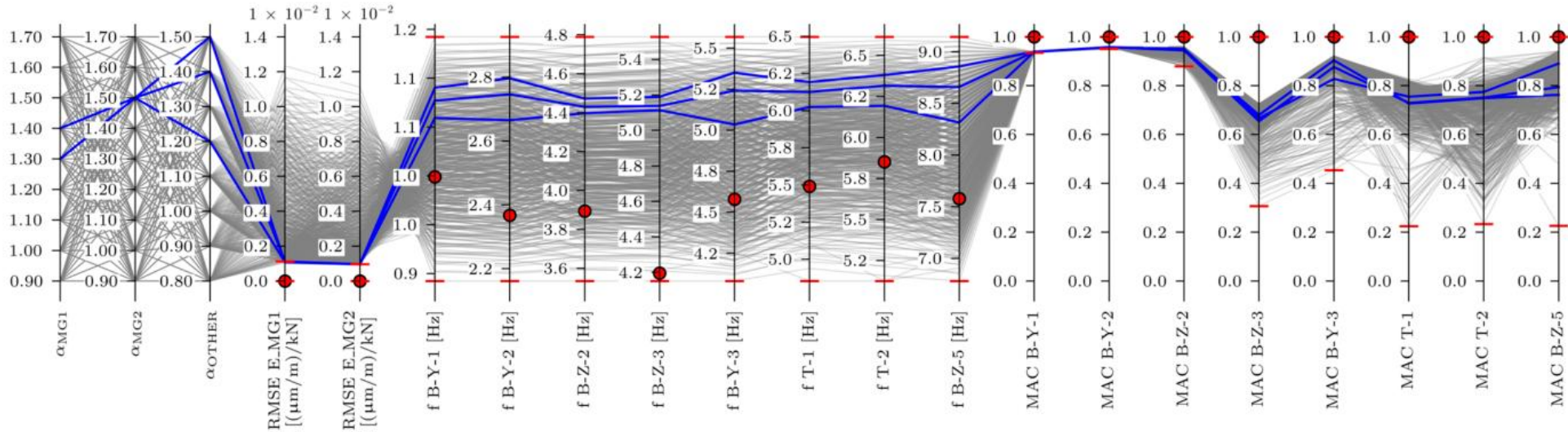


EDMF results

ACC-based approach

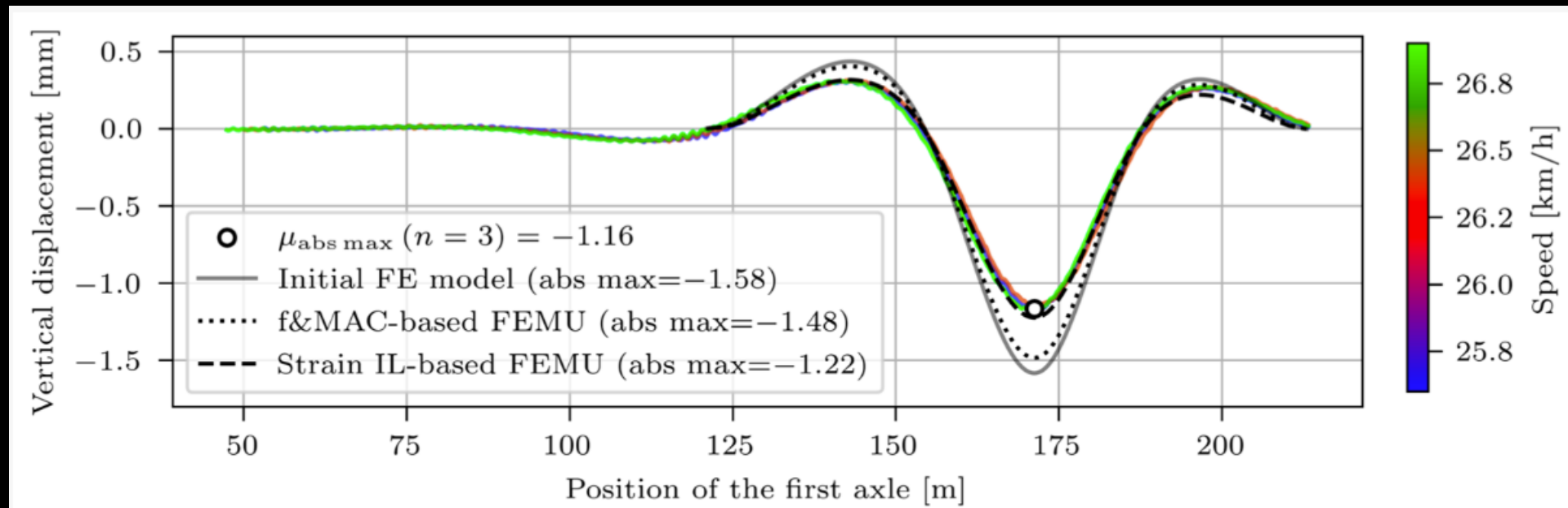


Strain-based (IL) approach (B-WIM)



Validation of Updated FE model

- up to 50% stiffness increase of the main girders
- updated boundary conditions (free → fixed)
- displacement prediction: IL-based > Acc-based



Take away

Define your GOAL'S

Execute proactive
management activities.

Use relevant data to
extract vital information